AlcorCut Pro — How it works

How some of the app's behind-the-scenes parts work: what is kept, what is done again, and why. Written for the app's own team and testers. Status of each part is noted (**built** or **planned**).

Sound in video exports: kept once, mixed fresh (planned)

Status: **planned, not built yet.** Goal: repeat audio-only exports in seconds, and faster video exports with sound. Sound processing is split into two layers.

1. Kept once per clip — the slow part

- Pulling the sound out of the video file. The sound sits in small slices between the picture data, so the phone has to read through the whole video file — gigabytes for 4K.
- The heavy processing: noise reduction, AI voice isolation, de-click, de-reverb.
- The result is kept in the video catalog, labelled with exactly what went into it: the file, the part used (the trim) and those heavy settings.

2. Done fresh on every export — the light part (your mix)

- EQ (more bass, more treble), volume, fades, pan, compressor.
- Music, extra tracks, ducking, audio transitions.
- These are quick sums for the processor, so they are never kept — always applied from the current settings.

What happens when you change something

You change…	What happens
Bass, treble, volume, fades, the mix	Nothing needs redoing — still seconds.
Pitch or speed of one clip	That clip's light layer is redone from its kept sound — still fast.
Noise reduction / voice isolation strength on a clip	Only that clip's heavy part is redone (slow for that one clip); every other clip stays instant.
Trim a clip longer	Only the new part is missing — that clip is redone.
Add a new clip	Only the new clip is processed.

So the sound can be tweaked freely: the editor always uses the latest settings, and only a clip changed in a heavy way costs time.

AI noise reduction in the video editor: done once per clip (built)

- On the first export, each clip with AI noise reduction is cleaned once, at full size, and kept in the video catalog ("AI noise reduction — clip 1 of 2 (done once, reused next time)").
- That export and every later one read the cleaned clip instead of running the AI on every frame again.
- Changing the strength cleans that clip again. **Slim down** in the catalog frees the cleaned clips (they are made again when needed).
- The cleaned clip is stored at a very high bitrate so it looks the same as the AI's output — a video can't practically be stored losslessly.
- Cloud AI (Topaz) is used instead when turned on; its results are kept per clip too, so they are never paid for twice.

AI noise reduction in the photo editor (built)

- Turning it on cleans the preview first (quick), then the full-size photo once in the background ("Cleaning the full-size photo").
- Both results are kept in the photo catalog, so reopening a photo or project never runs the AI again, and the export uses the full-size result.
- Clarity, texture, shadows, colour grading and the rest never make it run again. With RAW photos, exposure, white balance, highlights, shadows, whites and blacks do (the AI works on the developed picture) — so set those first. AI object removal and AI expand make a new picture and also need a new run.

Long video exports in parts (built)

- Films over 4 minutes are rendered in parts of about 2½ minutes, split only at plain cuts — never inside a picture or sound transition.
- Each part is rendered on its own; if one fails it is done again on its own instead of restarting the whole film.
- The parts are then joined into one video without re-encoding (only if they don't match is the join re-encoded).

Catalogs (built)

- **Photo catalogs** remember every photo you edit: a copy of the photo, its edit and settings, stars / flags / labels, and AI results.
- **Video catalogs** remember every clip you use: its last settings (given back when the clip is added to another project), proxies, stabilisation analysis, waveforms, thumbnails, render cache, cleaned AI clips, cloud AI results and transcripts.
- You choose how many photos / clips a catalog remembers (10 or more, or Unlimited); when it's full, the one used longest ago is let go first, with its files.
- Catalogs can be named, renamed, opened, deleted and moved (private storage, shared storage, SD card).
- Storage is lossless only: AI results as lossless PNG; RAW / TIFF copies packed losslessly when that saves space; JPEG, HEIC and video kept as they are.
- **Slim down** frees everything that can be made again and keeps photos, clips, copies and settings.

Photo export (built)

- Exports run in a separate background process with all its memory free for the photo, so full-resolution exports with AI tools fit.
- The size you choose is always the size saved. If memory still runs short, the heaviest steps run a little smaller and the photo is brought back to your size (the app says so).
- JPEG uses the app's own encoder: full colour detail (4:4:4); at 100 % quality hardly any compression. Ultra HDR JPEGs use the phone's encoder (the gain map lives inside its format).

Fast thumbnails in the video editor (built)

- Each clip shows a cover at once — the phone's own ready-made video thumbnail, or one quick frame.
- Filmstrips fill in coarse to fine (ends and middle first, then the gaps), several clips at a time, from the proxy when there is one.
- All thumbnails are kept in the video catalog, so a reopened project shows them straight away.

Chromecast (built)

- The phone encodes the picture (and sound) itself and serves it on the local Wi-Fi as a live stream; Google's free Cast library connects the TV to it.
- The TV runs about 3–5 seconds behind (Chromecast streams in short segments).
- Video editor: the viewer with titles, layers and the mixed sound of all tracks. Capture: the viewfinder with the microphone.
- Switching screens keeps the TV connected; it shows the AlcorCut Pro card in between.

Camera inside the editors (built)

- The Camera button in the photo and video editors opens the full Capture screen (every mode and pro control) over the editor, which stays open underneath.
- Photo editor: each photo goes straight into the filmstrip (without the automatic watermark). Video editor: videos and photos (as 5 s clips) go into the media list, not onto the timeline.